

Formula & Key Terms Booklet

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How to use this booklet

Every formula the competition is likely to need, grouped by topic, with each variable defined and a note on when to use it. Sections A to K hold the formulas; the two key-terms sections at the end decode acronyms and jargon. Worked examples for most formulas are in the Private Wealth & Portfolio Management Course.

- **Rates go in as decimals:** 5% is 0.05.
- **Periods match the rate:** use a monthly rate with monthly periods; formulas are annual unless stated.
- **Signs:** in spreadsheets, money paid out is negative and money received is positive.
- **Rules of thumb are labelled as such.** They are quick checks, not substitutes for the full formula.

Notation key

Symbol	Meaning
r	Rate of return, discount rate or interest rate per period
r_e, r_d	Cost of equity, cost of debt
r_f	Risk-free rate (T-bill or Treasury yield)
$E(R)$	Expected return
R_m, R_b, R_p	Return of the market, the benchmark, the portfolio
σ, σ^2	Standard deviation (volatility), variance
ρ	Correlation coefficient, from -1 to +1
Cov	Covariance
β	Beta: sensitivity to the market
α	Alpha: return beyond what risk explains
w	Portfolio weight
n, t	Number of periods; a given period
m	Compounding periods per year
g	Growth rate
π	Inflation rate
PV, FV	Present value, future value
PMT	Payment per period
C, F	Coupon payment, face (par) value of a bond
y	Yield to maturity
D	Dividend (in valuation) or duration (in bonds)
P	Price
E	Earnings (in valuation) or equity (in capital structure)
V	Firm value ($D + E$) or portfolio value
T	Tax rate (or time to expiry, in options)
b	Retention ratio = $1 - \text{payout ratio}$
A	Risk-aversion coefficient
$N(\cdot)$	Cumulative standard normal distribution
Δ	Change in
Σ, Π	Sum, product
$\ln, e$	Natural logarithm, Euler's number (about 2.718)

A • Time value of money and returns

A1 • Future and present value of a lump sum

$$FV = PV(1 + r)^n \quad PV = \frac{FV}{(1 + r)^n}$$

A2 • Compounding m times a year, and continuously

$$FV = PV \left(1 + \frac{r}{m}\right)^{mn} \quad FV = PV e^{rn}$$

A3 • Effective annual rate (EAR, also called APY)

$$EAR = \left(1 + \frac{r}{m}\right)^m - 1 \quad EAR_{\text{continuous}} = e^r - 1$$

Use it to compare rates quoted with different compounding.

A4 • Ordinary annuity (payments at the end of each period)

$$PV = PMT \times \frac{1 - (1 + r)^{-n}}{r} \quad FV = PMT \times \frac{(1 + r)^n - 1}{r}$$

For an annuity due (payments at the start of each period), multiply either result by $(1 + r)$.

A5 • Perpetuity and growing perpetuity

$$PV = \frac{PMT}{r} \quad PV = \frac{C_1}{r - g} \quad (r > g)$$

A6 • Growing annuity

$$PV = \frac{C_1}{r - g} \left[1 - \left(\frac{1 + g}{1 + r}\right)^n\right]$$

Values a stream that rises every year, such as withdrawals raised for inflation.

A7 • Loan or mortgage payment

$$PMT = PV \times \frac{r}{1 - (1 + r)^{-n}}$$

A8 • Net present value and internal rate of return

$$NPV = \sum_{t=0}^n \frac{CF_t}{(1 + r)^t} \quad 0 = \sum_{t=0}^n \frac{CF_t}{(1 + IRR)^t}$$

Accept a project when $NPV > 0$, or when its IRR exceeds the required return.

A9 • Holding period return

$$HPR = \frac{P_1 - P_0 + D_1}{P_0}$$

A10 • Arithmetic and geometric mean return

$$\bar{R} = \frac{1}{n} \sum_{t=1}^n R_t \quad R_G = \left[\prod_{t=1}^n (1 + R_t) \right]^{1/n} - 1$$

Use the geometric mean (CAGR) for past multi-year performance and the arithmetic mean for a single period's expected return.

A11 • Volatility drag

$$R_G \approx \bar{R} - \frac{\sigma^2}{2}$$

A12 • CAGR and annualizing

$$CAGR = \left(\frac{V_{\text{end}}}{V_{\text{begin}}} \right)^{1/n} - 1 \quad R_{\text{annual}} = (1 + R_{\text{period}})^k - 1$$

k is the number of periods per year (12 for monthly data). Do not annualize periods shorter than a year in client reports.

A13 • Real return (Fisher)

$$r_{\text{real}} = \frac{1 + r_{\text{nominal}}}{1 + \pi} - 1 \approx r_{\text{nominal}} - \pi$$

A14 • After-tax return on income taxed every year

$$r_{\text{after tax}} = r(1 - T)$$

Gains that are taxed only when you sell are handled in section I.

A15 · Log (continuously compounded) return

$$r_{\log} = \ln\left(\frac{P_1}{P_0}\right) = \ln(1 + HPR)$$

Log returns add across periods; simple returns compound.

A16 · Return on equity with leverage

$$R_E = R_A + \frac{D}{E}(R_A - r_d)$$

Borrowing magnifies gains when the asset return R_A beats the borrowing cost r_d , and magnifies losses when it does not.

A17 · Gain needed to recover from a loss L

$$\text{Required gain} = \frac{1}{1 - L} - 1$$

A 20% loss needs a 25% gain, a 33% loss needs 50%, and a 50% loss needs 100%.

A18 · Rule of 72 (rule of thumb)

Years to double $\approx 72 \div$ the rate in percent (69.3 for continuous compounding). Years to triple $\approx 114 \div$ the rate; to quadruple $\approx 144 \div$ the rate.

Time-weighted and money-weighted returns are in section D.

B • Risk and portfolio statistics

B1 • Variance and standard deviation

$$s^2 = \frac{\sum_{t=1}^n (R_t - \bar{R})^2}{n-1} \quad s = \sqrt{s^2}$$

Use $n-1$ for a sample of returns (STDEV.S) and n for a whole population (STDEV.P).

B2 • Annualizing volatility

$$\sigma_{\text{annual}} = \sigma_{\text{period}} \times \sqrt{k}$$

$k = 12$ for monthly data, 52 for weekly, about 252 for daily. Mean returns scale with k ; volatility scales with $\sqrt{k}$.

B3 • Covariance and correlation

$$\text{Cov}(X, Y) = \frac{\sum (X_t - \bar{X})(Y_t - \bar{Y})}{n-1} \quad \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

B4 • Beta

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\sigma_m^2} = \rho_{im} \frac{\sigma_i}{\sigma_m}$$

Beta is also the slope of a regression of the asset's returns on the market's (SLOPE in a spreadsheet).

B5 • Two-asset portfolio

$$E(R_p) = w_1 E(R_1) + w_2 E(R_2) \quad \sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

B6 • Portfolio of n assets (matrix form)

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}_{ij} = \mathbf{w}^\top \Sigma \mathbf{w}$$

Σ is the covariance matrix. Section K shows the spreadsheet version.

B7 • Portfolio beta

$$\beta_p = \sum_{i=1}^n w_i \beta_i$$

B8 • Equally weighted portfolio: why diversification has a floor

$$\sigma_p^2 = \frac{1}{n} \sigma^2 + \left(1 - \frac{1}{n}\right) \overline{\text{Cov}}$$

As n grows, the first term vanishes, leaving the average covariance: market risk cannot be diversified away.

B9 • Coefficient of variation

$$CV = \frac{\sigma}{\bar{R}}$$

Risk per unit of return; lower is better.

B10 • Downside deviation (target T)

$$\sigma_{\text{down}} = \sqrt{\frac{1}{n} \sum_{t=1}^n \min(R_t - T, 0)^2}$$

Counts only returns below the target; used in the Sortino ratio.

B11 • Skewness and excess kurtosis

$$\text{Skew} = \frac{1}{n} \sum \left(\frac{R_t - \bar{R}}{\sigma} \right)^3 \quad \text{Excess kurtosis} = \frac{1}{n} \sum \left(\frac{R_t - \bar{R}}{\sigma} \right)^4 - 3$$

Negative skew means occasional large losses; positive excess kurtosis means fat tails, so extreme moves happen more often than a bell curve predicts.

B12 • Z-score and the normal distribution

$$z = \frac{x - \mu}{\sigma}$$

Range around the mean	Share of outcomes inside (normal distribution)
$\pm 1 \sigma$	68.3%

Range around the mean	Share of outcomes inside (normal distribution)
$\pm 1.645 \sigma$	90% (5% in each tail)
$\pm 1.96 \sigma$	95%
$\pm 2.33 \sigma$	98% (1% in each tail)
$\pm 2.58 \sigma$	99%

B13 · Confidence interval for a mean

$$\bar{x} \pm z \times \frac{s}{\sqrt{n}}$$

B14 · Maximum drawdown

$$MDD = \max_t \left(\frac{\text{Peak before } t - V_t}{\text{Peak before } t} \right)$$

The largest peak-to-trough fall over the period.

B15 · Parametric value at risk (VaR)

$$VaR_\alpha = (z_\alpha \sigma - \mu) \times V \quad VaR_{h \text{ periods}} = (z_\alpha \sigma \sqrt{h} - \mu h) \times V$$

$z = 1.645$ for 95% and 2.33 for 99% confidence. VaR is the loss not exceeded with that confidence; it says nothing about the size of losses beyond it.

B16 · Conditional VaR (expected shortfall), normal distribution

$$ES_\alpha = \left(\sigma \frac{\varphi(z_\alpha)}{1 - \alpha} - \mu \right) \times V$$

φ is the standard normal density. At 95% confidence, $\varphi(1.645) \div 0.05 \approx 2.06$, so $ES \approx (2.06\sigma - \mu) \times V$. It is the average loss in the worst 5% of cases.

B17 · Roy's safety-first ratio

$$SFR = \frac{E(R_p) - R_L}{\sigma_p} \quad P(R_p < R_L) \approx N(-SFR)$$

R_L is the minimum acceptable return. Choose the portfolio with the highest SFR to minimize the chance of falling short.

B18 · Tracking error and R^2

$$TE = \sigma(R_p - R_b) \quad R^2 = \rho^2 \text{ (one-factor regression)}$$

R^2 is the share of a fund's return variance explained by its benchmark or factors.

B19 · Concentration: Herfindahl–Hirschman index

$$HHI = \sum_{i=1}^n w_i^2 \quad N_{\text{effective}} = \frac{1}{HHI}$$

Twenty equal positions give $HHI = 0.05$, the same as 20 effective holdings. Uneven weights reduce the effective number.

B20 · Risk contribution

$$MCR_i = \frac{(\Sigma \mathbf{w})_i}{\sigma_p} \quad RC_i = w_i \times MCR_i \quad \sum_i RC_i = \sigma_p$$

$RC_i \div \sigma_p$ is each holding's percentage share of total risk. A sleeve can hold a third of the money and most of the risk.

B21 · Diversification ratio

$$DR = \frac{\sum_i w_i \sigma_i}{\sigma_p}$$

Above 1 means diversification is reducing risk; higher is more diversified.

B22 · Linear regression (single factor)

$$R_{i,t} = \alpha + \beta R_{m,t} + \varepsilon_t \quad \beta = \frac{Cov(R_i, R_m)}{Var(R_m)} \quad \alpha = \bar{R}_i - \beta \bar{R}_m$$

The t-statistic of a coefficient is the estimate $\div$ its standard error; above about 2 is usually treated as statistically significant.

C · Portfolio theory, CAPM and factor models

C1 · Minimum-variance weight for two assets

$$w_1^* = \frac{\sigma_2^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2} \quad w_2^* = 1 - w_1^*$$

C2 · Capital allocation line (risky portfolio P plus the risk-free asset)

$$E(R_C) = r_f + y[E(R_P) - r_f] \quad \sigma_C = y\sigma_P$$

y is the share in the risky portfolio. The line's slope is P's Sharpe ratio, and the tangency portfolio maximizes it.

C3 · Investor utility and the optimal risky share

$$U = E(r) - \frac{1}{2}A\sigma^2 \quad y^* = \frac{E(R_P) - r_f}{A\sigma_P^2}$$

A higher risk-aversion coefficient A means a smaller risky share. Typical values of A run from about 2 (tolerant) to 6 or more (very averse).

C4 · Capital market line

$$E(R_p) = r_f + \frac{E(R_m) - r_f}{\sigma_m} \sigma_p$$

For efficient portfolios only; uses total risk σ .

C5 · CAPM and the security market line

$$E(R_i) = r_f + \beta_i[E(R_m) - r_f]$$

For any asset; uses systematic risk β . $E(R_m) - r_f$ is the market risk premium.

C6 · Jensen's alpha

$$\alpha_p = R_p - [r_f + \beta_p(R_m - r_f)]$$

C7 · Single-index model: total risk split in two

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma^2(\varepsilon_i)$$

Systematic risk plus firm-specific risk. Only the first is rewarded, because diversification removes the second.

C8 · Fama–French and Carhart factor models

$$R_i - r_f = \alpha_i + \beta_i(R_m - r_f) + s_i SMB + h_i HML + m_i UMD + \varepsilon_i$$

The three-factor model uses market, size (SMB) and value (HML). Carhart adds momentum (UMD). The five-factor model replaces momentum with profitability (RMW) and investment (CMA).

C9 · Arbitrage pricing theory

$$E(R_i) = r_f + \beta_{i1}\lambda_1 + \beta_{i2}\lambda_2 + \dots + \beta_{ik}\lambda_k$$

λ_k is the risk premium for factor k, such as unexpected inflation or GDP growth.

C10 · Grinold–Kroner expected stock return

$$E(R) \approx \frac{D}{P} + i + g - \Delta S + \Delta \frac{P}{E}$$

Dividend yield + expected inflation i + real earnings growth g – the yearly change in shares outstanding (buybacks make ΔS negative and add to return) + the yearly change in the P/E ratio.

C11 · Building-block expected bond return

$$E(R_{bond}) \approx y_0 - \text{expected credit losses} \pm \text{roll-down and rate changes}$$

For high-quality bonds, today's yield is the best single estimate of the next 5–10 years' return.

C12 · Implied equilibrium returns (Black–Litterman starting point)

$$\Pi = \lambda \Sigma \mathbf{w}_{mkt} \quad \lambda = \frac{E(R_m) - r_f}{\sigma_m^2}$$

The returns that make market-cap weights optimal. Black–Litterman then blends in your views, weighted by your confidence.

C13 · Inverse-volatility (naive risk parity) weights

$$w_i = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}$$

Full risk parity equalizes each asset's risk contribution (B20) instead: $RC_i = \sigma_p \div n$.

C14 · Fundamental law of active management

$$IR \approx IC \times \sqrt{BR}$$

The information ratio depends on skill (information coefficient IC, the correlation between forecasts and outcomes) and breadth (BR, the number of independent bets per year).

D • Risk-adjusted performance and attribution

D1 • Sharpe ratio

$$S_p = \frac{R_p - r_f}{\sigma_p}$$

Excess return per unit of total risk. For reference, US stocks from 1928 to 2023 scored about 0.43 on annual data (average excess return over T-bills of 8.3 points ÷ 19.6% volatility).

D2 • Sortino ratio

$$\text{Sortino} = \frac{R_p - T}{\sigma_{\text{down}}}$$

Like Sharpe, but penalizes only volatility below the target T (B10). It suits loss-averse clients.

D3 • Treynor ratio

$$T_p = \frac{R_p - r_f}{\beta_p}$$

Excess return per unit of market risk. Use it for a sleeve that sits inside a diversified portfolio.

D4 • Information ratio

$$IR = \frac{R_p - R_b}{\sigma(R_p - R_b)} = \frac{\text{Active return}}{\text{Tracking error}}$$

Skill relative to a benchmark. Above 0.5 sustained over years is strong.

D5 • M² (Modigliani risk-adjusted performance)

$$M^2 = r_f + S_p \sigma_b \quad M^2 \text{ excess} = M^2 - R_b$$

The return the portfolio would have earned at the benchmark's volatility, in percent.

D6 • Calmar ratio

$$\text{Calmar} = \frac{\text{Annualized return}}{|\text{Maximum drawdown}|}$$

D7 • Upside and downside capture

$$\text{Up capture} = \frac{R_p \text{ in up periods}}{R_b \text{ in up periods}} \quad \text{Down capture} = \frac{R_p \text{ in down periods}}{R_b \text{ in down periods}}$$

A conservative client wants down capture well below 100%. Capture ratio = up ÷ down; above 1 is good.

D8 • Active return and active share

$$\text{Active return} = R_p - R_b \quad \text{Active share} = \frac{1}{2} \sum_i |w_{p,i} - w_{b,i}|$$

Active share runs from 0% (an index fund) to 100% (no overlap with the benchmark).

D9 • Time-weighted return

$$TWR = \left[\prod_{i=1}^n (1 + r_i) \right]^{1/\text{years}} - 1 \quad r_i = \frac{V_{\text{end},i} - V_{\text{begin},i}}{V_{\text{begin},i}}$$

Split the period at every external cash flow; each sub-period's beginning value includes that flow. It removes the effect of the client's deposit timing and measures the manager.

D10 • Modified Dietz return (approximate TWR for one period)

$$R = \frac{EMV - BMV - \sum CF_i}{BMV + \sum w_i CF_i} \quad w_i = \frac{\text{days remaining after flow } i}{\text{days in period}}$$

D11 • Money-weighted return

$$\sum_{t=0}^n \frac{CF_t}{(1 + MWR)^t} = 0$$

The IRR of all deposits, withdrawals and the ending value (XIRR in a spreadsheet). It measures the client's own experience.

D12 • Brinson attribution (segment i)

$$\text{Allocation}_i = (w_{p,i} - w_{b,i}) R_{b,i} \quad \text{Selection}_i = w_{b,i} (R_{p,i} - R_{b,i}) \quad \text{Interaction}_i = (w_{p,i} - w_{b,i})(R_{p,i} - R_{b,i})$$

The three effects summed over all segments equal $R_p - R_b$. The Brinson–Fachler variant measures allocation against the total benchmark return:

$$\text{Allocation}_i^{BF} = (w_{p,i} - w_{b,i})(R_{b,i} - R_b)$$

D13 · Linking returns across periods

$$R_{\text{cumulative}} = (1 + R_1)(1 + R_2) \cdots (1 + R_n) - 1$$

D14 · Fee drag over n years

$$\frac{FV_{\text{after fees}}}{FV_{\text{before fees}}} = \left(\frac{1 + r - f}{1 + r} \right)^n$$

At 6% gross, a 1% fee leaves about 75% of the ending wealth after 30 years.

D15 · Hit rate (batting average)

$$\text{Hit rate} = \frac{\text{Periods beating the benchmark}}{\text{Total periods}}$$

E · Bonds, yields, duration and TIPS

E1 · Bond price

$$P = \sum_{t=1}^n \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n} \quad \text{semiannual: } C \rightarrow \frac{C}{2}, y \rightarrow \frac{y}{2}, n \rightarrow 2n$$

E2 · Zero-coupon bond

$$P = \frac{F}{(1+y)^n} \quad y = \left(\frac{F}{P}\right)^{1/n} - 1$$

E3 · Current yield and approximate yield to maturity

$$\text{Current yield} = \frac{C}{P} \quad YTM \approx \frac{C + (F - P)/n}{(F + P)/2}$$

The exact YTM solves E1 for y (YIELD or RATE in a spreadsheet). Yield to call uses the call date and call price instead of maturity and par; yield to worst is the lowest of them.

E4 · Bond-equivalent and effective annual yield

$$BEY = 2 \times y_{\text{semiannual}} \quad EAY = \left(1 + \frac{BEY}{2}\right)^2 - 1$$

E5 · Treasury bill yields (t = days to maturity)

$$\text{Discount yield} = \frac{F - P}{F} \times \frac{360}{t} \quad \text{Investment (BEY) yield} = \frac{F - P}{P} \times \frac{365}{t}$$

Bills are quoted on the discount basis; the investment yield is the better comparison with other bonds.

E6 · Accrued interest and dirty price

$$AI = \frac{C}{m} \times \frac{\text{days since last coupon}}{\text{days in coupon period}} \quad \text{Dirty price} = \text{Clean price} + AI$$

E7 · Spot and forward rates

$$(1 + s_n)^n = (1 + s_m)^m (1 + f_{m,n-m})^{n-m}$$

Example: $(1 + s_2)^2 = (1 + s_1)(1 + f_{1,1})$. The forward rate f is the rate the curve implies for the future period.

E8 · Macaulay and modified duration

$$D_{Mac} = \frac{1}{P} \sum_{t=1}^n \frac{t \times CF_t}{(1+y)^t} \quad D_{Mod} = \frac{D_{Mac}}{1 + y/m}$$

A zero-coupon bond's Macaulay duration equals its maturity; a perpetuity's is $(1 + y) \div y$.

E9 · Effective duration and convexity (for bonds with options)

$$D_{Eff} = \frac{P_- - P_+}{2 P_0 \Delta y} \quad C_{Eff} = \frac{P_- + P_+ - 2P_0}{P_0 (\Delta y)^2}$$

P_- and P_+ are the prices after yields fall and rise by Δy .

E10 · Price change from a yield change

$$\frac{\Delta P}{P} \approx -D_{Mod} \Delta y + \frac{1}{2} C (\Delta y)^2$$

Rule of thumb: a 1-point rise in yields cuts a bond's price by about its duration in percent.

E11 · DV01 and portfolio duration

$$DV01 = D_{Mod} \times P \times 0.0001 \quad D_p = \sum_i w_i D_i$$

DV01 (or PV01) is the dollar change in value for a 1-basis-point move in yield.

E12 · Expected return on a bond over a horizon

$$E(R) \approx y + \text{roll-down} - D_{Mod} \Delta y + \frac{1}{2} C (\Delta y)^2 - \text{credit losses} \pm \text{currency}$$

E13 · Credit risk

$$EL = PD \times LGD \times EAD \quad LGD = 1 - \text{recovery rate} \quad \text{Spread} \approx PD \times LGD$$

Expected loss from the probability of default (PD), the loss given default (LGD) and the exposure at default (EAD). The spread approximation gives the annual compensation needed just to cover expected losses.

E14 · Taxable-equivalent yield

$$TEY = \frac{y_{muni}}{1 - t} \quad t_{combined} \approx t_{federal} + t_{state}$$

Use the combined rate for an in-state bond that is exempt from both federal and state tax.

E15 · TIPS mechanics

$$\text{Index ratio} = \frac{CPI_{ref, today}}{CPI_{ref, issue}} \quad \text{Adjusted principal} = \text{Par} \times \text{Index ratio}$$

$$\text{Interest payment} = \frac{\text{Real coupon}}{2} \times \text{Adjusted principal} \quad \text{Paid at maturity} = \max(\text{Adjusted principal}, \text{Par})$$

E16 · Breakeven inflation

$$\pi_{BE} = y_{nominal} - y_{real} \quad \text{exact: } \pi_{BE} = \frac{1 + y_{nominal}}{1 + y_{real}} - 1$$

If realized inflation beats the breakeven rate, TIPS outperform nominal Treasuries of the same maturity.

E17 · I bond composite rate

$$r_{composite} = r_{fixed} + 2\pi_{semiannual} + r_{fixed} \times \pi_{semiannual}$$

E18 · Bond holding period return

$$HPR = \frac{P_1 - P_0 + C}{P_0}$$

E19 · Bond-fund horizon (rule of thumb)

A bond fund kept at a constant duration tends to return close to its starting yield over a horizon of roughly $2 \times \text{duration} - 1$ years, because higher reinvestment rates slowly offset the price loss from rising rates.

F · Equity valuation: DCF, DDM, WACC, multiples

F1 · Gordon growth dividend discount model

$$P_0 = \frac{D_1}{r - g} = \frac{D_0(1 + g)}{r - g} \quad r = \frac{D_1}{P_0} + g \quad g_{\text{implied}} = r - \frac{D_1}{P_0}$$

Valid only when $r > g$ and growth is stable forever.

F2 · Two-stage dividend discount model

$$P_0 = \sum_{t=1}^n \frac{D_t}{(1 + r)^t} + \frac{1}{(1 + r)^n} \times \frac{D_{n+1}}{r - g_L}$$

High growth for n years, then long-run growth g_L .

F3 · H-model (growth fading in a straight line)

$$P_0 = \frac{D_0(1 + g_L)}{r - g_L} + \frac{D_0 H (g_S - g_L)}{r - g_L}$$

g_S is today's short-term growth, g_L the long-run rate, and H half the length of the fade period in years.

F4 · Present value of growth opportunities

$$P_0 = \frac{E_1}{r} + PVGO$$

The price of a no-growth firm plus the value the market places on future growth.

F5 · Free cash flow to the firm (FCFF)

$$FCFF = EBIT(1 - T) + D\&A - \text{Capex} - \Delta NWC = CFO + \text{Interest}(1 - T) - \text{Capex}$$

F6 · Free cash flow to equity (FCFE)

$$FCFE = FCFF - \text{Interest}(1 - T) + \text{Net borrowing} = CFO - \text{Capex} + \text{Net borrowing}$$

Discount FCFF at WACC to get firm value; discount FCFE at the cost of equity to get equity value. Never mix them.

F7 · DCF value and terminal value

$$V_0 = \sum_{t=1}^n \frac{FCFF_t}{(1 + WACC)^t} + \frac{TV_n}{(1 + WACC)^n} \quad TV_n = \frac{FCFF_n(1 + g)}{WACC - g} \quad \text{or} \quad TV_n = \text{Multiple} \times \text{Metric}_n$$

F8 · Enterprise value and the bridge to equity

$$EV = \text{Market cap} + \text{Debt} + \text{Preferred} + \text{Minority interest} - \text{Cash}$$

$$\text{Value per share} = \frac{EV - \text{Net debt} - \text{Preferred} - \text{Minority interest}}{\text{Diluted shares}}$$

F9 · Weighted average cost of capital

$$WACC = \frac{E}{V} r_e + \frac{D}{V} r_d(1 - T) + \frac{P}{V} r_p$$

Use market-value weights; P is preferred stock.

F10 · Cost of equity

$$r_e = r_f + \beta(ERP) \quad r_e = r_f + ERP + \text{size premium} + \text{specific-risk premium}$$

CAPM, then the build-up method used for small or private companies. A quick cross-check: the company's bond yield plus 3–5 points (rule of thumb).

F11 · Levering and unlevering beta (Hamada) and adjusted beta (Blume)

$$\beta_L = \beta_U \left[1 + (1 - T) \frac{D}{E} \right] \quad \beta_{\text{adjusted}} = \frac{2}{3} \beta_{\text{raw}} + \frac{1}{3}$$

Unlever peers' betas, average them, then relever at your company's debt-to-equity ratio.

F12 · Justified multiples from fundamentals

$$\frac{P_0}{E_1} = \frac{1 - b}{r - g} \quad \frac{P_0}{E_0} = \frac{(1 - b)(1 + g)}{r - g} \quad \frac{P_0}{B_0} = \frac{ROE - g}{r - g} \quad \frac{P_0}{S_0} = \frac{PM(1 - b)(1 + g)}{r - g}$$

b is the retention ratio and PM the net profit margin. A company whose ROE equals its cost of equity deserves a P/B of 1.

F13 · Other multiples and yields

$$PEG = \frac{P/E}{g \times 100} \quad \text{Earnings yield} = \frac{E}{P} \quad \text{FCF yield} = \frac{FCF}{\text{Market cap}} \quad \text{Shareholder yield} = \frac{\text{Dividends} + \text{Net buyback}}{\text{Market cap}}$$

F14 • Residual income model

$$RI_t = E_t - r B_{t-1} = (ROE_t - r) B_{t-1} \quad V_0 = B_0 + \sum_{t=1}^{\infty} \frac{RI_t}{(1+r)^t}$$

F15 • NOPAT, economic profit and EVA

$$NOPAT = EBIT(1 - T) \quad EVA = NOPAT - WACC \times IC = (ROIC - WACC) \times IC$$

IC is invested capital (debt + equity – cash).

F16 • Value driver formula

$$V_0 = \frac{NOPAT_1 \left(1 - \frac{g}{RONIC}\right)}{WACC - g}$$

RONIC is the return on new invested capital. Growth adds value only when RONIC > WACC.

F17 • Sustainable growth

$$g = ROE \times b \quad g_{NOPAT} = \text{Reinvestment rate} \times ROIC$$

F18 • Margin of safety and upside

$$\text{Margin of safety} = \frac{V - P}{V} \quad \text{Upside} = \frac{V}{P} - 1$$

F19 • Graham number (rule of thumb)

$$V_{Graham} = \sqrt{22.5 \times EPS \times BVPS}$$

Benjamin Graham's ceiling for a defensive investor: a P/E of 15 times a P/B of 1.5. A crude screen, not a valuation.

G · Financial ratios and business analytics

Average balance-sheet figures (beginning + end ÷ 2) are preferred when a ratio divides a flow by a stock. Compare every ratio with the company's own history and with peers.

G1 · Profitability

Ratio	Formula
Gross margin	Gross profit ÷ revenue
Operating (EBIT) margin	EBIT ÷ revenue
EBITDA margin	EBITDA ÷ revenue
Net margin	Net income ÷ revenue
Return on assets (ROA)	Net income ÷ average total assets
Return on equity (ROE)	Net income ÷ average shareholders' equity
Return on invested capital (ROIC)	NOPAT ÷ average invested capital (debt + equity – cash)
Incremental ROIC	Change in NOPAT ÷ change in invested capital

G2 · Efficiency and working capital

Ratio	Formula
Asset turnover	Revenue ÷ average total assets
Inventory turnover	Cost of goods sold ÷ average inventory
Days inventory outstanding (DIO)	365 ÷ inventory turnover
Receivables turnover	Revenue ÷ average receivables
Days sales outstanding (DSO)	365 ÷ receivables turnover
Payables turnover	Cost of goods sold ÷ average payables
Days payables outstanding (DPO)	365 ÷ payables turnover
Cash conversion cycle	DIO + DSO – DPO

G3 · Liquidity

Ratio	Formula
Current ratio	Current assets ÷ current liabilities
Quick (acid-test) ratio	(Cash + marketable securities + receivables) ÷ current liabilities
Cash ratio	(Cash + marketable securities) ÷ current liabilities

G4 · Leverage and coverage

Ratio	Formula
Debt to equity	Total debt ÷ total equity
Debt to capital	Total debt ÷ (debt + equity)
Net debt to EBITDA	(Debt – cash) ÷ EBITDA
Equity multiplier	Average total assets ÷ average equity
Interest coverage	EBIT ÷ interest expense
Fixed-charge coverage	(EBIT + lease payments) ÷ (interest + lease payments)
FFO to debt (credit analysts)	Funds from operations ÷ total debt

G5 · Cash flow and payout

Ratio	Formula
Free cash flow	Operating cash flow – capital expenditure
FCF margin	FCF ÷ revenue
FCF conversion	FCF ÷ net income
Accruals ratio	(Net income – operating cash flow) ÷ average total assets
Capex intensity	Capex ÷ revenue; capex ÷ depreciation above 1 signals growth investment
Payout ratio	Dividends ÷ net income (or ÷ FCF)
Retention ratio (b)	1 – payout ratio
Dividend coverage	Net income (or FCF) ÷ dividends
Reinvestment rate	(Capex – D&A + increase in NWC) ÷ NOPAT

G6 · Per share

Measure	Formula
Diluted EPS	(Net income – preferred dividends) ÷ diluted weighted-average shares
Book value per share	Common equity ÷ shares outstanding
Dividends per share	Total common dividends ÷ shares outstanding
FCF per share	FCF ÷ diluted shares

G7 · DuPont analysis

$$ROE = \underbrace{\frac{NI}{Sales}}_{\text{margin}} \times \underbrace{\frac{Sales}{Assets}}_{\text{turnover}} \times \underbrace{\frac{Assets}{Equity}}_{\text{leverage}}$$

$$ROE = \frac{NI}{EBT} \times \frac{EBT}{EBIT} \times \frac{EBIT}{Sales} \times \frac{Sales}{Assets} \times \frac{Assets}{Equity}$$

The five-step version adds the tax burden (NI ÷ EBT) and the interest burden (EBT ÷ EBIT).

G8 · Operating, financial and total leverage

$$DOL = \frac{\% \Delta EBIT}{\% \Delta Sales} = \frac{\text{Contribution margin}}{EBIT} \quad DFL = \frac{\% \Delta EPS}{\% \Delta EBIT} = \frac{EBIT}{EBIT - \text{Interest}} \quad DTL = DOL \times DFL$$

G9 · Break-even

$$Q_{BE} = \frac{\text{Fixed costs}}{P - VC} \quad \text{Break-even revenue} = \frac{\text{Fixed costs}}{\text{Contribution margin ratio}} \quad CM \text{ ratio} = \frac{P - VC}{P}$$

P is the price per unit and VC the variable cost per unit.

G10 · Altman Z-score (public manufacturers)

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

X_1 = working capital ÷ total assets; X_2 = retained earnings ÷ total assets; X_3 = EBIT ÷ total assets; X_4 = market value of equity ÷ total liabilities; X_5 = sales ÷ total assets. Above 2.99 is the safe zone, 1.81–2.99 the grey zone and below 1.81 the distress zone.

G11 · Piotroski F-score (0 to 9)

Score 1 point for each test passed; 8–9 signals strength, 0–2 weakness.

Area	Tests
Profitability	ROA positive; operating cash flow positive; ROA higher than last year; operating cash flow above net income
Leverage and liquidity	Long-term debt ratio lower than last year; current ratio higher; no new shares issued
Efficiency	Gross margin higher than last year; asset turnover higher

G12 · Beneish M-score (earnings manipulation screen)

$$M = -4.84 + 0.920 DSRI + 0.528 GMI + 0.404 AQI + 0.892 SGI + 0.115 DEPI - 0.172 SGAI + 4.679 TATA - 0.327$$

Each input is a year-on-year index: days' sales in receivables (DSRI), gross margin (GMI), asset quality (AQI), sales growth (SGI), depreciation (DEPI), SG&A (SGAI), total accruals to total assets (TATA) and leverage (LVGI). A score above about –1.78 flags a possible manipulator. Use it as a prompt for questions, not as proof.

G13 · Industry metrics

Industry	Metric	Formula
Software	Rule of 40	Revenue growth % + FCF margin %; 40 or more is healthy
Software	Net revenue retention	(Starting recurring revenue + expansion – contraction – churn) ÷ starting recurring revenue
Subscription	Customer lifetime value (LTV)	Revenue per customer × gross margin ÷ churn rate
Subscription	CAC payback (months)	Customer acquisition cost ÷ (monthly revenue per customer × gross margin)
Banks	Net interest margin	(Interest income – interest expense) ÷ average earning assets
Banks	Efficiency ratio	Non-interest expense ÷ revenue; lower is better
Banks	CET1 ratio	Common equity tier 1 capital ÷ risk-weighted assets
Insurance	Combined ratio	Loss ratio + expense ratio; below 100% is an underwriting profit
REITs	Funds from operations (FFO)	Net income + real-estate depreciation and amortization – gains on property sales

Industry	Metric	Formula
REITs	Adjusted FFO	FFO – recurring capex – straight-line rent adjustments
Real estate	Cap rate	Net operating income ÷ property value
Retail	Same-store sales growth	Sales growth at stores open at least a year
Airlines	Load factor	Revenue passenger miles ÷ available seat miles
Semiconductors	Book-to-bill	Orders received ÷ units shipped and billed; above 1 means demand is growing

G14 · Analytics shortcuts

$$\text{YoY growth} = \frac{X_t}{X_{t-1}} - 1 \quad \text{Common-size line} = \frac{\text{Line item}}{\text{Revenue (or total assets)}}$$

H · Economics and macroeconomics

H1 · GDP by expenditure

$$GDP = C + I + G + (X - M)$$

Consumption, investment, government spending and net exports.

H2 · Real GDP and the GDP deflator

$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{Deflator}/100} \quad \text{Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

H3 · Growth accounting (Solow)

$$g_Y = g_A + \alpha g_K + (1 - \alpha) g_L$$

Output growth = productivity growth + capital share × capital growth + labor share × labor growth.

H4 · Output gap

$$\text{Output gap} = \frac{\text{Actual GDP} - \text{Potential GDP}}{\text{Potential GDP}}$$

Positive means the economy is running hot (inflation pressure); negative means slack.

H5 · Inflation and the CPI

$$\pi_t = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} \quad CPI = \frac{\text{Cost of basket today}}{\text{Cost of basket in base year}} \times 100$$

H6 · Converting to real (today's) dollars

$$\text{Value in today's dollars} = \text{Past value} \times \frac{CPI_{\text{today}}}{CPI_{\text{then}}}$$

H7 · Labor market

$$\text{Unemployment rate} = \frac{\text{Unemployed}}{\text{Labor force}} \quad \text{Participation rate} = \frac{\text{Labor force}}{\text{Working-age population}}$$

H8 · Sahm rule recession indicator

$$S_t = \bar{u}_{3m,t} - \min(\bar{u}_{3m} \text{ over the prior 12 months})$$

A reading of 0.5 points or more has signaled the start of past US recessions.

H9 · Okun's law (rule of thumb)

$$\Delta u \approx -0.5 \times (g_{\text{real GDP}} - g_{\text{potential}})$$

Growth 1 point below potential tends to raise unemployment by about half a point.

H10 · Phillips curve (expectations-augmented)

$$\pi = \pi^e - \beta(u - u^*) + \text{supply shocks}$$

u^* is the natural rate of unemployment.

H11 · Taylor rule

$$i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5(y - y^*)$$

r^* is the neutral real rate, π^* the 2% target and $y - y^*$ the output gap in percent. It is a benchmark for where the policy rate "should" be, not what the Fed must do.

H12 · Fisher equation and the real interest rate

$$(1 + i) = (1 + r)(1 + \pi^e) \quad r \approx i - \pi^e$$

H13 · Quantity theory of money

$$MV = PY \quad \% \Delta M + \% \Delta V \approx \% \Delta P + \% \Delta Y$$

Money supply × velocity = price level × real output.

H14 · Money multiplier (simple)

$$\text{Money multiplier} = \frac{1}{\text{Reserve ratio}}$$

A textbook upper bound; in practice banks' lending is constrained by capital and demand.

H15 · Fiscal multipliers

- - - -

$$k = \frac{1}{1 - MPC} = \frac{1}{MPS} \quad k_{\text{tax}} = \frac{-MPC}{1 - MPC} \quad k_{\text{open economy}} = \frac{1}{1 - MPC(1 - t) + MPM}$$

$MPC + MPS = 1$; t is the tax rate and MPM the marginal propensity to import. A balanced-budget change has a multiplier of 1.

H16 · Government debt dynamics

$$\Delta d \approx (r - g)d - s$$

d is debt ÷ GDP, r the real interest rate on the debt, g real GDP growth and s the primary surplus ÷ GDP. When r exceeds g , the debt ratio rises unless the government runs a primary surplus.

H17 · Elasticities

$$E_d = \frac{\% \Delta Q_d}{\% \Delta P} \quad E_{\text{midpoint}} = \frac{(Q_2 - Q_1) / [(Q_1 + Q_2)/2]}{(P_2 - P_1) / [(P_1 + P_2)/2]}$$

Income elasticity = $\% \Delta Q \div \% \Delta \text{income}$; cross-price elasticity = $\% \Delta Q$ of good $x \div \% \Delta P$ of good y . When $|E_d| > 1$ demand is elastic, and cutting price raises total revenue; when $|E_d| < 1$ the firm has pricing power.

H18 · Revenue, cost and profit

$$TR = P \times Q \quad MR = \frac{\Delta TR}{\Delta Q} \quad MC = \frac{\Delta TC}{\Delta Q} \quad \text{Profit is maximized where } MR = MC$$

H19 · Market concentration

$$HHI = \sum_i s_i^2 \quad CR_4 = s_1 + s_2 + s_3 + s_4$$

With market shares s in percent, HHI runs from near 0 (fragmented) to 10,000 (monopoly). Regulators treat markets in the high thousands as highly concentrated; check the current merger guidelines for thresholds.

H20 · Exchange rates and parity conditions

$$\frac{F}{S} = \frac{1 + i_d}{1 + i_f} \quad E(\% \Delta S) \approx i_d - i_f \quad \% \Delta S \approx \pi_d - \pi_f$$

S and F are the spot and forward prices of one unit of foreign currency in domestic currency. The first is covered interest parity (a no-arbitrage condition), the second uncovered interest parity, and the third relative purchasing power parity; the last two hold only loosely and over long periods.

H21 · Real exchange rate and home-currency return

$$\text{Real exchange rate} = S \times \frac{P_f}{P_d} \quad R_{\text{home}} = (1 + R_{\text{local}})(1 + R_{FX}) - 1$$

H22 · Trade and the balance of payments

$$\text{Terms of trade} = \frac{\text{Export price index}}{\text{Import price index}} \times 100 \quad CA + KA + FA = 0$$

The current, capital and financial accounts sum to zero (net of errors and omissions).

H23 · Rule of 70 (doubling time for any growth rate)

$$\text{Years to double} \approx \frac{70}{g \text{ in percent}}$$

At 2% real growth, an economy doubles in size in about 35 years.

I · Personal finance, retirement and tax

I1 · Household balance sheet and cash flow

$$\text{Net worth} = \text{Assets} - \text{Liabilities} \quad \text{Savings rate} = \frac{\text{Annual savings}}{\text{Annual income}} \quad DTI = \frac{\text{Monthly debt payments}}{\text{Gross monthly income}}$$

Lenders generally want a low debt-to-income ratio (DTI); many cap it somewhere around 36–45%.

I2 · Emergency fund (rule of thumb)

$$\text{Emergency fund} = \text{Monthly essential expenses} \times (3 \text{ to } 6)$$

Use 6–12 months for irregular incomes and retirees.

I3 · Cost of a goal in future dollars

$$\text{Future cost} = \text{Cost today} \times (1 + \pi)^n$$

I4 · Savings needed to reach a goal

$$PMT = [FV_{\text{goal}} - PV(1 + r)^n] \times \frac{r}{(1 + r)^n - 1}$$

I5 · Required return

$$FV_{\text{goal}} = PV(1 + r)^n + PMT \frac{(1 + r)^n - 1}{r} \quad \text{solve for } r$$

Use RATE in a spreadsheet, or Goal Seek when there are withdrawals along the way.

I6 · Portfolio needed for retirement

$$\text{Annual gap} = \text{Spending} - \text{Social Security} - \text{Pensions} \quad \text{Portfolio needed} \approx \frac{\text{Annual gap}}{\text{Withdrawal rate}}$$

At a 4% withdrawal rate, that is 25 times the annual gap. Planners often assume spending of 70–85% of pre-retirement income (the replacement ratio).

I7 · Sustainable withdrawal (annuity method)

$$W = PV \times \frac{r}{1 - (1 + r)^{-n}} \quad W_{\text{real}}: \text{ use } r_{\text{real}} = \frac{1 + r}{1 + \pi} - 1$$

The level withdrawal that exhausts the portfolio in exactly n years. Using the real rate gives withdrawals that keep pace with inflation.

I8 · Human capital (simplified)

$$HC_0 = \sum_{t=1}^N \frac{\text{Expected earnings}_t}{(1 + r_f + \text{risk premium})^t}$$

Add a larger risk premium for less stable careers.

I9 · Life insurance need (income-replacement method)

$$\text{Need} \approx PV(\text{family's lost income and goals}) - \text{Existing assets and coverage}$$

I10 · Tax rates

$$\text{Effective rate} = \frac{\text{Total tax}}{\text{Income}} \quad \text{Marginal rate} = \text{tax on the next dollar}$$

Use the marginal rate for investment decisions.

I11 · Capital gains tax

$$\text{Tax} = (P_{\text{sale}} - \text{Basis}) \times t_{cg} \quad \text{After-tax proceeds} = P_{\text{sale}} - (P_{\text{sale}} - \text{Basis}) t_{cg}$$

I12 · Accrual vs deferred taxation

$$FV_{\text{taxed yearly}} = PV [1 + r(1 - t)]^n \quad FV_{\text{taxed at sale}} = PV [(1 + r)^n(1 - t) + t]$$

The second formula taxes only the gain at the end (basis = PV). Deferral lets untaxed gains compound.

I13 · Traditional vs Roth accounts

$$FV_{\text{traditional, after tax}} = C(1 + r)^n(1 - t_{\text{withdrawal}}) \quad FV_{\text{Roth}} = C(1 - t_{\text{contribution}})(1 + r)^n$$

C is the pretax amount available to save. The two are equal when the tax rates today and at withdrawal are equal, so the choice turns on whether you expect your tax rate to rise or fall.

I14 · Required minimum distribution

$$RMD = \frac{\text{Prior year-end account balance}}{\text{Distribution period (IRS Uniform Lifetime Table)}}$$

RMDs from traditional IRAs start at 73, rising to 75 for people born in 1960 or later.

I15 · Social Security delayed retirement credits

Benefit at claim age $\approx$ Full benefit $\times (1 + 0.08 \times \text{years delayed past full retirement age})$

Credits stop at age 70. With a full retirement age of 67, waiting until 70 raises the benefit by about 24%.

I16 · US rules worth knowing

Rule	What it says
Long-term capital gains	Assets held more than a year: 0%, 15% or 20% federal rates depending on income
Net investment income tax	An extra 3.8% for higher earners
Short-term gains and interest	Taxed at ordinary income rates
Qualified dividends	Taxed at long-term capital-gains rates if the holding-period test is met
Collectibles, including physically backed gold ETFs	Long-term gains taxed at up to 28%
Capital losses	Offset gains, plus up to \$3,000 a year of ordinary income; the rest carries forward
Wash-sale rule	A loss is disallowed if a substantially identical security is bought within 30 days before or after the sale
Early withdrawals	Withdrawals from retirement accounts before 59½ generally incur a 10% penalty, with exceptions
Bank deposits	FDIC insurance covers up to \$250,000 per depositor, per insured bank, per ownership category
Brokerage accounts	SIPC protects up to \$500,000 per customer, including up to \$250,000 of cash, if a broker fails (not against market losses)

Income thresholds change every year; check the IRS for the current figures.

I17 · 2026 contribution limits

Account	2026 limit
401(k), 403(b), most 457 plans and the Thrift Savings Plan	\$24,500
Catch-up, age 50 and over	\$8,000
Catch-up, ages 60–63	\$11,250
Traditional or Roth IRA	\$7,500
IRA catch-up, age 50 and over	\$1,100
SIMPLE IRA	\$17,000, plus a \$4,000 catch-up
Roth IRA income phase-out, single filers	\$153,000–\$168,000
Roth IRA income phase-out, married filing jointly	\$242,000–\$252,000

Source: [InvestmentNews](#), citing the IRS.

J · Options and derivatives essentials

S is the stock price, K the strike price, c and p the call and put premiums, T the time to expiry in years and r the risk-free rate.

J1 · Payoffs at expiry

$$\text{Call} = \max(S_T - K, 0) \quad \text{Put} = \max(K - S_T, 0)$$

The buyer's profit is the payoff minus the premium; the seller's profit is the premium minus the payoff.

J2 · Intrinsic and time value

$$\text{Call intrinsic} = \max(S - K, 0) \quad \text{Put intrinsic} = \max(K - S, 0) \quad \text{Time value} = \text{Premium} - \text{Intrinsic value}$$

J3 · Breakeven prices for option buyers

$$\text{Long call: } S_T = K + c \quad \text{Long put: } S_T = K - p$$

J4 · Put-call parity (European options)

$$c + Ke^{-rT} = p + S_0$$

With known dividends, replace S_0 with S_0 minus the present value of the dividends.

J5 · Protective put (own the stock, buy a put)

$$\text{Maximum loss} = S_0 - K + p \quad \text{Breakeven} = S_0 + p$$

Insurance with unlimited upside, paid for by the premium.

J6 · Covered call (own the stock, sell a call)

$$\text{Maximum gain} = K - S_0 + c \quad \text{Maximum loss} = S_0 - c \quad \text{Breakeven} = S_0 - c$$

Income now in exchange for capped upside.

J7 · Collar (own the stock, buy a put at K_1 , sell a call at K_2)

$$\text{Floor} = K_1 - S_0 - p + c \quad \text{Cap} = K_2 - S_0 - p + c$$

Selling the call pays for some or all of the put. A "zero-cost collar" sets the strikes so that $c = p$.

J8 · Black-Scholes-Merton

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2) \quad p = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \quad d_2 = d_1 - \sigma\sqrt{T}$$

For a stock with dividend yield q, replace S_0 with $S_0 e^{(-qT)}$ and $r + \sigma^2/2$ with $r - q + \sigma^2/2$.

J9 · The Greeks

Greek	Measures	Value or sign for a long option
Delta	Price change per \$1 move in the stock	Call $N(d_1)$, between 0 and 1; put $N(d_1) - 1$, between -1 and 0
Gamma	Change in delta per \$1 move	Positive; highest near the strike close to expiry
Theta	Value lost per day as time passes	Usually negative
Vega	Value change per 1 -point change in implied volatility	Positive
Rho	Value change per 1 -point change in interest rates	Positive for calls, negative for puts

J10 · Delta hedging

$$\text{Options needed to hedge} = \frac{\text{Shares held}}{\text{Delta per option}}$$

J11 · Forward and futures pricing (cost of carry)

$$F_0 = S_0 e^{(r-q)T} \quad F_0 = S_0 e^{(r+u-y)T}$$

The first is for an asset with income yield q; the second for a commodity with storage cost u and convenience yield y. Currency forwards follow covered interest parity (section H).

J12 · Changing a portfolio's beta with index futures

$$N_f = \frac{(\beta_{\text{target}} - \beta_{\text{portfolio}}) \times V_p}{F \times \text{multiplier}}$$

A negative N_f means selling contracts. A target beta of 0 fully hedges market risk.

J13 · Minimum-variance hedge ratio

$$h^* = \rho \frac{\sigma_S}{\sigma_F} \quad N^* = h^* \frac{Q_A}{Q_F}$$

σ_S and σ_F are the volatilities of the spot and futures price changes; Q_A is the position size and Q_F the size of one contract.

J14 · Commodity futures return

$$R_{\text{futures index}} \approx \text{Spot return} + \text{Roll yield} + \text{Collateral return}$$

Roll yield is negative in contango (later contracts cost more) and positive in backwardation.

K · Spreadsheet functions and rules of thumb

K1 · Excel and Google Sheets functions

Rates and periods must match (a monthly rate with monthly periods). Money paid out is negative.

Function	What it returns	Example
FV(rate, nper, pmt, [pv], [type])	Future value	=FV(5%, 10, 0, -100000) gives \$162,889
PV(rate, nper, pmt, [fv], [type])	Present value	=PV(8%, 5, 0, -1000)
PMT(rate, nper, pv, [fv], [type])	Payment per period	=PMT(6%/12, 360, -400000) for a mortgage
NPER(rate, pmt, pv, [fv])	Number of periods	Years to reach a goal
RATE(nper, pmt, pv, [fv])	Rate per period	Required return for a goal
NPV(rate, values)	Present value of values starting one period from now	Add the time-0 cash flow separately
XNPV(rate, values, dates)	NPV with exact dates	DCF with irregular dates
IRR(values), XIRR(values, dates)	Internal rate of return	XIRR gives the money-weighted return
EFFECT(rate, npery), NOMINAL(rate, npery)	Effective or nominal annual rate	=EFFECT(6%, 12) gives 6.17%
AVERAGE, MEDIAN, GEOMEAN	Means	=GEOMEAN(1 + returns) - 1 for the geometric mean
STDEV.S, VAR.S	Sample standard deviation and variance	Monthly volatility × SQRT(12) to annualize
CORREL(x, y), COVARIANCE.S(x, y)	Correlation, covariance	Inputs for portfolio risk
SLOPE(y, x), INTERCEPT(y, x), RSQ(y, x)	Beta, alpha, R ² of a regression	=SLOPE(stock returns, market returns)
LINEST(y, x-range, TRUE, TRUE)	Multiple regression with statistics	Factor regressions
SUMPRODUCT(weights, values)	Weighted sum	Portfolio expected return or duration
MMULT, TRANSPOSE	Matrix maths	Portfolio volatility (below)
NORM.S.INV(p), NORM.INV(p, mean, sd), NORM.DIST(x, mean, sd, TRUE)	Normal distribution	=NORM.S.INV(0.95) gives 1.645
RAND()	Uniform random number	Monte Carlo draws (below)
PRICE, YIELD, DURATION, MDURATION	Bond price, YTM, Macaulay and modified duration	Need settlement and maturity dates
TBILLPRICE, TBILLYIELD, TBILLEQ	T-bill price, yield and bond-equivalent yield	Comparing bills with notes
ACCRINT	Accrued interest	Dirty price
XLOOKUP, INDEX with MATCH, SUMIFS	Lookups and conditional sums	Building comparison tables
GOOGLEFINANCE(ticker, attribute, start, end, interval)	Prices and history (Google Sheets only)	=GOOGLEFINANCE("VTI", "close", DATE(2021,1,1), TODAY(), "DAILY")
Data Table (What-If Analysis)	A grid of results for two changing inputs	DCF sensitivity to WACC and growth
Goal Seek	The input that hits a target	Implied growth in a reverse DCF
Solver add-in	Optimization with constraints	Minimum-variance weights

Portfolio volatility from weights w (a column) and covariance matrix Cov:

```
=SQRT(MMULT(MMULT(TRANSPOSE(w), Cov), w))
```

One Monte Carlo year of returns (copy across years and down 1,000+ rows):

```
=NORM.INV(RAND(), expected_return, volatility)
```

Linking a column of periodic returns into a cumulative return:

```
=PRODUCT(1 + returns) - 1
```

Enter it as an array formula in older versions of Excel.

K2 · Rules of thumb

Rule	What it says	Caveat
Rule of 72	Years to double $\approx 72 \div \text{rate}$; triple $\approx 114 \div \text{rate}$; quadruple $\approx 144 \div \text{rate}$	Approximate; least accurate at high rates
Duration rule	A 1-point rise in yields cuts a bond's price by about its duration in percent	Ignores convexity on large moves
Bond-fund horizon	Return $\approx$ starting yield over about $2 \times \text{duration} - 1$ years	Assumes a constant duration and no defaults
Loss recovery	-20% needs +25%, -33% needs +50%, -50% needs +100%	Exact: $1 \div (1 - \text{loss}) - 1$
4% rule	Withdraw 4% of the starting portfolio, then raise it with inflation; historically lasted 30 years in the US	Not a guarantee; depends on returns, fees and horizon
25× rule	Portfolio needed $\approx 25 \times$ annual spending drawn from it	The inverse of the 4% rule

Rule	What it says	Caveat
Emergency fund	3–6 months of essential expenses	6–12 months for irregular income or retirees
"110 minus age"	Rough share in stocks	Ignores the actual client; a starting point only
Diversification	20–30 stocks across sectors remove most company - specific risk	Not market risk
Position limits	Common IPS limits: 5% per stock, 25% per sector	Set them from the client's risk profile
Fees	Over 30 years at 6% gross, a 1% fee takes about a quarter of the ending wealth	Exact: section D14
Losing years	US stocks lost money in 26 of 96 calendar years from 1928 to 2023, about one in four	Losses cluster in recessions
Bear markets	The S&P 500 has entered a bear market about every 3.5 years on average since 1929	Timing is unpredictable
Sharpe ratio	US stocks scored about 0.43 on annual data, 1928–2023	Higher is better; compare like with like
Lump sum vs averaging in	Investing at once beat dollar - cost averaging about two - thirds of the time in Vanguard's research	Averaging in can still help a nervous client commit
Basis points	100 basis points = 1 percentage point	"Up 25 bp" means up 0.25 points
Annualizing volatility	Monthly $\times \sqrt{12}$; daily $\times \sqrt{252}$	Assumes independent returns

Key terms, part 1: acronyms A to Z

The acronyms you will meet in filings, research, the news and judges' questions, each with what it stands for and what it means.

Filings, forms and accounts

Acronym	Stands for	What it means
10-K	Annual report filed with the SEC	Audited financial statements, business description, risk factors and MD&A
10-Q	Quarterly report	Unaudited quarterly financial statements
8-K	Current report	Filed after material events: earnings, deals, executive departures
13F	Institutional holdings report	Quarterly list of US stocks held by managers with more than \$100 million in such securities
20-F	Annual report of a foreign issuer	The 10-K equivalent for non-US companies listed in the US
DEF 14A	Definitive proxy statement	Executive pay, board members, shareholder votes
Form 4	Statement of changes in beneficial ownership	Filed when insiders buy or sell their company's shares
S-1	Registration statement	The disclosure document filed before an IPO
401(k), 403(b), 457	Sections of the US tax code	Employer retirement plans for private, nonprofit and government employees
529	Section 529 plan	Tax-advantaged education savings account
K-1	Schedule K-1	Tax form from a partnership; some commodity funds issue one

A-B

Acronym	Stands for	What it means
ABS	Asset-backed security	A bond backed by a pool of loans such as car loans or credit-card balances
ACWI	All Country World Index (MSCI)	Global stock index covering developed and emerging markets
ADR	American depositary receipt	US-traded certificate representing shares of a foreign company
ADV	Average daily volume	Shares traded per day; a liquidity gauge
AFFO	Adjusted funds from operations	FFO minus recurring capex; a REIT's cash available for dividends
ALM	Asset-liability management	Matching assets to future obligations
AML	Anti-money laundering	Rules to detect and stop the laundering of illicit funds
AP	Authorized participant	A dealer that creates and redeems ETF shares
APR	Annual percentage rate	Yearly interest rate before compounding
APT	Arbitrage pricing theory	A multi-factor model of expected returns
APY	Annual percentage yield	Yearly rate including compounding (same as EAR)
ARPU	Average revenue per user	Revenue ÷ users or subscribers
ARR	Annual recurring revenue	Yearly value of subscription contracts
ATH	All-time high	The highest price ever reached
AUM	Assets under management	Money a manager or fund oversees
BDC	Business development company	Listed vehicle that lends to or invests in mid-sized private companies
BEY	Bond-equivalent yield	Semiannual yield × 2
BHB	Brinson, Hood and Beebower	Authors of the 1986 asset-allocation study and the classic attribution model
BoE, BoJ	Bank of England, Bank of Japan	Central banks of the UK and Japan
bp, bps	Basis point(s)	0.01 percentage point; 100 bp = 1%
BVPS	Book value per share	Common equity ÷ shares outstanding

C-D

Acronym	Stands for	What it means
CAC	Customer acquisition cost	Sales and marketing spend per new customer
CAGR	Compound annual growth rate	The smoothed yearly growth rate between two values
CAL	Capital allocation line	Risk-return mixes of a risky portfolio and the risk-free asset
CAPE	Cyclically adjusted price-to-earnings ratio	Price ÷ 10-year average inflation-adjusted earnings (Shiller P/E)
CAPM	Capital asset pricing model	Expected return = risk-free rate + beta × market risk premium
Capex	Capital expenditure	Spending on long-lived assets such as plants and equipment
Cboe	Chicago Board Options Exchange	Options exchange; publishes the VIX
CCC	Cash conversion cycle	DSO + DIO – DPO (also a junk credit rating)
CD	Certificate of deposit	Fixed-term bank deposit
CDO, CLO	Collateralized debt or loan obligation	Securities carved from pools of bonds or loans
CDS	Credit default swap	Contract that pays out if a borrower defaults

Acronym	Stands for	What it means
CEF	Closed-end fund	Fund with a fixed number of shares that trades at a premium or discount to NAV
CET1	Common equity tier 1	A bank's core capital as a share of risk-weighted assets
CFA	Chartered Financial Analyst	Investment credential awarded by CFA Institute
CFO, CFI, CFF	Cash flow from operations, investing, financing	The three sections of the cash flow statement (CFO also means chief financial officer)
CFP	Certified Financial Planner	Financial-planning credential
CIO	Chief investment officer	Head of investment decisions
CMA	Capital market assumptions	Forecast returns, risks and correlations (also the "conservative minus aggressive" factor)
CMBS	Commercial mortgage-backed security	Bond backed by commercial property loans
CML	Capital market line	The CAL that uses the market portfolio
COGS	Cost of goods sold	Direct costs of the products sold
CPI, CPI-U	Consumer price index (for all urban consumers)	The main US inflation measure; CPI-U drives TIPS and I bonds
CR4	Four-firm concentration ratio	Combined market share of the four largest firms
CTA	Commodity trading advisor	Manager of futures strategies, often trend-following
CV	Coefficient of variation	Standard deviation ÷ mean
CVaR	Conditional value at risk	Average loss beyond the VaR level (expected shortfall)
D&A	Depreciation and amortization	Non-cash charges that spread asset costs over time
D/E	Debt to equity	Total debt ÷ shareholders' equity
DCA	Dollar-cost averaging	Investing equal amounts at regular intervals
DCF	Discounted cash flow	Valuing an asset as the present value of future cash flows
DDM	Dividend discount model	Valuing a stock as the present value of future dividends
DIO, DSO, DPO	Days inventory, sales and payables outstanding	Working-capital cycle measures
DJIA	Dow Jones Industrial Average	Price-weighted index of 30 large US stocks
DOL, DFL, DTL	Degree of operating, financial and total leverage	How much profits swing with sales
DPI	Distributions to paid-in capital	Cash returned ÷ cash invested in a private fund
DPS	Dividends per share	Total dividends ÷ shares
DRIP	Dividend reinvestment plan	Automatically buys more shares with dividends
DTI	Debt to income	Monthly debt payments ÷ gross monthly income
DV01	Dollar value of a basis point	Price change for a 0.01% yield change (same as PV01)

E–F

Acronym	Stands for	What it means
EAFF	Europe, Australasia and Far East	MSCI index of developed markets outside North America
EAR	Effective annual rate	The yearly rate after compounding
EBIT	Earnings before interest and taxes	Operating profit
EBITDA	Earnings before interest, taxes, depreciation and amortization	A proxy for operating cash profit
EBT	Earnings before taxes	Pretax income
ECB	European Central Bank	Central bank of the euro area
EDGAR	Electronic Data Gathering, Analysis and Retrieval	The SEC's free database of filings
EM	Emerging markets	Developing economies such as China, India and Brazil
EMH	Efficient market hypothesis	Prices reflect available information
EPS	Earnings per share	Net income ÷ shares
ERP	Equity risk premium	Extra return investors demand for owning stocks over risk-free bonds
ES	Expected shortfall	Same as CVaR
ESG	Environmental, social and governance	Non-financial factors in investment analysis
ETF	Exchange-traded fund	A fund traded on an exchange like a stock
ETN	Exchange-traded note	Unsecured debt that tracks an index; carries the issuer's credit risk
EV	Enterprise value	Market cap + debt + preferred + minority interest – cash
EVA	Economic value added	NOPAT – WACC × invested capital
FCF, FCFE, FCFF	Free cash flow (to equity, to the firm)	Cash left after reinvestment, for shareholders or for all capital providers
FDIC	Federal Deposit Insurance Corporation	Insures US bank deposits up to \$250,000 per depositor, per bank, per ownership category
Fed	Federal Reserve	The US central bank
FFO	Funds from operations	REIT earnings measure: net income + real-estate depreciation – gains on sales
FFR	Federal funds rate	Overnight rate between banks that the Fed targets
FINRA	Financial Industry Regulatory Authority	Self-regulator of US broker-dealers

Acronym	Stands for	What it means
FOMC	Federal Open Market Committee	The Fed committee that sets interest - rate policy
FOMO	Fear of missing out	The urge to chase rising assets
FRA	Full retirement age	Age for unreduced Social Security benefits (also forward rate agreement)
FRED	Federal Reserve Economic Data	The St. Louis Fed's free economic database
FRN	Floating - rate note	Bond whose coupon resets with short - term rates
FX	Foreign exchange	Currencies
FY	Fiscal year	A company's 12 - month reporting year

G-I

Acronym	Stands for	What it means
GAAP	Generally accepted accounting principles	US accounting rules
GARP	Growth at a reasonable price	Style blending growth and value
GDP	Gross domestic product	Value of everything an economy produces
GICS	Global Industry Classification Standard	The 11 - sector system used by S&P and MSCI
GIPS	Global Investment Performance Standards	Rules for fair performance reporting
GNI	Gross national income	GDP plus net income from abroad
GO bond	General obligation bond	Municipal bond backed by the issuer's taxing power
GP, LP	General partner, limited partner	Manager and investors of a private fund
HFT	High - frequency trading	Computerized trading in microseconds
HHI	Herfindahl-Hirschman index	Sum of squared shares; measures concentration
HML	High minus low	The value factor
HNWI, UHNWI	(Ultra -)high - net - worth individual	Wealthy private - banking clients
HPR	Holding period return	Return over the time an asset is held
HSA	Health savings account	Tax - advantaged account for medical costs
HY	High yield	Bonds rated below investment grade
IC	Information coefficient	Correlation between forecasts and outcomes (also investment committee)
IFRS	International Financial Reporting Standards	Accounting rules used in most countries outside the US
IG	Investment grade	Bonds rated BBB - / Baa3 or higher
IMF	International Monetary Fund	International lender and economic monitor
IORB	Interest on reserve balances	Rate the Fed pays banks on reserves
IPO	Initial public offering	A company's first sale of shares to the public
IPS	Investment Policy Statement	Document setting a client's objectives, constraints and rules
IR	Information ratio	Active return ÷ tracking error (also investor relations)
IRA	Individual retirement account	Tax - advantaged personal retirement account
IRR	Internal rate of return	Discount rate that sets NPV to zero
IRS	Internal Revenue Service	The US tax authority
ISM	Institute for Supply Management	Publishes the manufacturing and services PMIs
ISSB	International Sustainability Standards Board	Sets global sustainability disclosure standards
ITM, ATM, OTM	In, at, out of the money	Whether an option has intrinsic value

K-M

Acronym	Stands for	What it means
KPI	Key performance indicator	A measure management tracks
KYC	Know your customer	Rules to verify client identity and understand their profile
LBO	Leveraged buyout	Acquisition funded mostly with debt
LDI	Liability - driven investing	Investing to fund known future liabilities
LEI	Leading Economic Index	Conference Board composite of leading indicators
LGD	Loss given default	Share of an exposure lost when a borrower defaults
LIBOR	London Interbank Offered Rate	Former benchmark rate, replaced by SOFR in the US in 2023
LTCG, STCG	Long - term and short - term capital gains	Gains on assets held more than, or at most, one year
LTM, TTM	Last or trailing twelve months	The most recent four quarters combined
LTV	Lifetime value (of a customer)	Also loan - to - value for mortgages
M&A	Mergers and acquisitions	Companies combining or buying each other
M ²	Modigliani-Modigliani measure	Risk - adjusted return at the benchmark's volatility

Acronym	Stands for	What it means
MBS	Mortgage-backed security	Bond backed by home loans
MD&A	Management's discussion and analysis	Narrative section of the 10-K
MDD	Maximum drawdown	Largest peak-to-trough fall
MLP	Master limited partnership	Listed partnership, often in energy pipelines
MMF	Money market fund	Fund holding very short-term debt
MOIC	Multiple on invested capital	Total value ÷ money invested
MPC, MPS	Marginal propensity to consume, to save	Share of extra income spent or saved
MPT	Modern portfolio theory	Markowitz's mean-variance framework
MSCI	(Formerly Morgan Stanley Capital International)	Major index provider
MVO	Mean-variance optimization	Choosing weights for the best return per unit of risk
MWR	Money-weighted return	IRR of the client's cash flows
MYGA	Multi-year guaranteed annuity	Fixed-rate annuity, like an insurer's CD

N-P

Acronym	Stands for	What it means
NAV	Net asset value	Assets minus liabilities, per fund share
Nareit	National Association of Real Estate Investment Trusts	REIT trade group and index provider
NBER	National Bureau of Economic Research	Dates the start and end of US recessions
NFP	Nonfarm payrolls	Monthly US jobs count
NI	Net income	Bottom-line profit
NII	Net interest income	Interest earned minus interest paid (banks)
NIIT	Net investment income tax	Extra 3.8% US tax on investment income for higher earners
NIM	Net interest margin	NII ÷ average earning assets
NOI	Net operating income	Property income after operating costs
NOPAT	Net operating profit after tax	EBIT × (1 – tax rate)
NPL	Non-performing loan	Loan in or near default
NPV	Net present value	Present value of inflows minus outflows
NRR	Net revenue retention	Revenue kept and expanded from existing customers
NWC	Net working capital	Current assets minus current liabilities
NYSE	New York Stock Exchange	The largest US stock exchange
OAS	Option-adjusted spread	Credit spread after removing embedded options
OCF	Operating cash flow	Same as CFO
OECD	Organisation for Economic Co-operation and Development	Club of advanced economies; publishes data and forecasts
OKR	Objectives and key results	Goal-setting framework
OPEC+	Organization of the Petroleum Exporting Countries plus allies	Oil producers coordinating output
OTC	Over the counter	Traded between dealers rather than on an exchange
P&L	Profit and loss	The income statement, or gains and losses on a position
P/B, P/E, P/S	Price to book, earnings, sales	Valuation multiples
PCE	Personal consumption expenditures	Price index behind the Fed's 2% inflation target
PD	Probability of default	Chance a borrower defaults
PE	Private equity	Ownership stakes in private companies (also P/E)
PEG	Price/earnings-to-growth	P/E ÷ growth rate
PESTEL	Political, economic, social, technological, environmental, legal	Framework for external forces
PMI	Purchasing managers' index	Business survey; above 50 means expansion
PP&E	Property, plant and equipment	Physical long-term assets
PPI	Producer price index	Prices received by producers
PPP	Purchasing power parity	Exchange rates adjust to equalize price levels
PV, PV01	Present value; price value of a basis point	Today's value of future cash; same as DV01
PVGO	Present value of growth opportunities	Part of a stock's price owed to future growth

Q-S

Acronym	Stands for	What it means
QE, QT	Quantitative easing, tightening	Central-bank bond buying or shrinking of holdings
QoQ	Quarter over quarter	Change from the previous quarter

Acronym	Stands for	What it means
QP	Qualified purchaser	Investor with at least \$5 million in investments
R&D	Research and development	Spending on new products
r*	Neutral real interest rate	Rate that neither stimulates nor restrains the economy
REIT	Real estate investment trust	Company that owns income property and pays out most of its income
RIA	Registered investment adviser	Adviser registered with the SEC or a state; owes a fiduciary duty
RMD	Required minimum distribution	Mandatory yearly withdrawal from tax - deferred accounts
RMW	Robust minus weak	The profitability factor
ROA, ROE, ROI, ROIC	Return on assets, equity, investment, invested capital	Profitability measures
RONIC	Return on new invested capital	Return earned on fresh investment
RRTTLLU	Return, risk, time horizon, taxes, liquidity, legal, unique circumstances	The IPS framework
RSU	Restricted stock unit	Share grant that vests over time
S&P	Standard & Poor's	S&P Dow Jones Indices (index provider) and S&P Global Ratings
SAA, TAA	Strategic, tactical asset allocation	Long - term policy mix; short - term tilts within ranges
SAAR	Seasonally adjusted annual rate	How many US economic data are quoted
SaaS	Software as a service	Subscription software
SAMURAI	Specified, appropriate, measurable, unambiguous, reflective, accountable, investable	Qualities of a valid benchmark
SASB	Sustainability Accounting Standards Board	Industry ESG standards, now maintained by the ISSB
SBC	Stock - based compensation	Pay in shares or options; a real cost
SEC	Securities and Exchange Commission	US securities regulator
SEP	Summary of Economic Projections	Fed officials' forecasts, including the dot plot
SG&A	Selling, general and administrative expenses	Overhead costs
SIPC	Securities Investor Protection Corporation	Protects brokerage customers up to \$500,000 (including \$250,000 cash) if a broker fails
SMA	Separately managed account	Portfolio owned directly by the client (also simple moving average)
SMB	Small minus big	The size factor
SML	Security market line	CAPM line of expected return against beta
SOFR	Secured Overnight Financing Rate	Benchmark rate that replaced USD LIBOR
SOTP	Sum of the parts	Valuing each business segment separately
SPAC	Special purpose acquisition company	Listed shell that raises cash to buy a private company
SPIA	Single premium immediate annuity	Lump sum exchanged for lifetime income
SPIVA	S&P Indices Versus Active	Scorecards comparing active funds with indexes
STRIPS	Separate Trading of Registered Interest and Principal of Securities	Zero - coupon pieces of Treasuries
SWOT	Strengths, weaknesses, opportunities, threats	Strategy framework

T-Z

Acronym	Stands for	What it means
T+1	Trade date plus one business day	US settlement cycle for stocks and ETFs
TAM, SAM, SOM	Total addressable, serviceable available, serviceable obtainable market	Layers of market sizing
TE	Tracking error	Volatility of returns relative to the benchmark
TER	Total expense ratio	All annual fund costs as a percentage of assets
TEY	Taxable - equivalent yield	Muni yield ÷ (1 - tax rate)
TIPS	Treasury Inflation - Protected Securities	Treasuries whose principal tracks CPI
TLH	Tax - loss harvesting	Selling losers to offset taxable gains
TR	Total return	Price change plus income
TSP	Thrift Savings Plan	The 401(k) - style plan for US federal employees
TVM	Time value of money	A dollar today is worth more than a dollar later
TVPI	Total value to paid - in capital	Private fund's value plus distributions ÷ capital invested
TWR	Time - weighted return	Return that removes the effect of cash - flow timing
UIT	Unit investment trust	Fund with a fixed portfolio and end date
UMA	Unified managed account	One account holding several managed strategies
UMD	Up minus down	The momentum factor
VaR	Value at risk	Loss not exceeded at a given confidence level
VC	Venture capital	Investment in early - stage companies

Acronym	Stands for	What it means
VIX	Cboe Volatility Index	Expected 30-day S&P 500 volatility from option prices; the "fear gauge"
VRIO	Valuable, rare, inimitable, organized	Test for lasting competitive advantage
WACC	Weighted average cost of capital	Blended required return of debt and equity investors
WAM	Weighted average maturity	Average time to maturity of a bond portfolio
WInS	Wharton Investment Simulator	The competition's trading platform
WTI	West Texas Intermediate	US crude-oil benchmark
YoY	Year over year	Change from the same period a year earlier
YTC, YTM, YTW	Yield to call, maturity, worst	Bond return measures to different end dates
YTD	Year to date	From January 1 to today
ZIRP	Zero interest-rate policy	Policy rates held near zero

Key terms, part 2: concepts and jargon by theme

Plain-English definitions of the words that appear in research, client meetings and judges' questions.

Markets and trading

Term	Meaning
Bull market	A sustained rise in prices, often defined as 20% or more from a low
Bear market	A fall of 20% or more from a peak
Correction	A fall of 10–20% from a recent high
Rally, sell - off	A sharp rise; a sharp, broad fall
Volatility	How much prices swing, usually measured by standard deviation
Liquidity	How quickly an asset can be sold close to its fair value
Bid, ask, spread	The best price buyers pay, the best price sellers accept, and the gap between them
Market, limit and stop orders	Fill now at any price; fill only at your price or better; trigger a market order when a price is hit
Short selling	Selling borrowed shares in the hope of buying them back cheaper
Margin	Money borrowed from a broker to buy securities
Leverage	Using debt or derivatives to magnify exposure, and so gains and losses
Market capitalization	Share price × shares outstanding
Free float	Shares available for public trading
Index, benchmark	A basket that measures a market; the yardstick a portfolio is judged against
Blue chip	A large, established, financially sound company
Ex - dividend date	The first day a buyer no longer receives the upcoming dividend
Stock split	More shares at a proportionally lower price; total value unchanged
Buyback	A company repurchasing its own shares
Dilution	New shares shrinking existing owners' percentage stake
Capitulation	Panicked, high - volume selling, often near a market bottom
Dead - cat bounce	A brief rebound in a falling market
Circuit breaker	An automatic trading halt after a large market drop
Settlement	The transfer of cash and securities after a trade (T+1 in the US)

Investing styles and strategies

Term	Meaning
Value investing	Buying assets priced below their estimated intrinsic value
Growth investing	Buying companies expected to grow faster than average
GARP	Growth at a reasonable price: a blend of growth and value
Quality investing	Favoring profitable, low - debt, stable companies
Momentum	Buying recent winners and avoiding recent losers
Income investing	Focusing on dividends and interest
Dividend growth investing	Owning companies with long records of raising dividends
Contrarian	Going against the crowd
Passive investing	Tracking an index at low cost
Active management	Trying to beat a benchmark through selection or timing
Factor (smart - beta) investing	Tilting toward characteristics linked to excess returns or lower risk
Core - satellite	A low - cost index core plus targeted active or thematic satellites
Buy and hold	Owning for the long run and trading rarely
Sector rotation	Shifting between sectors as the business cycle turns
Thematic investing	Concentrating on a trend such as clean energy or robotics
ESG and impact investing	Including environmental, social and governance factors; targeting measurable positive outcomes
Dollar - cost averaging	Investing equal amounts at regular intervals
Asset location	Choosing which account (taxable, deferred, Roth) holds each asset
Tax - loss harvesting	Realizing losses to offset taxable gains

Bonds and interest rates

Term	Meaning
Par (face) value	The amount repaid at maturity, usually \$1,000 or \$100

Term	Meaning
Coupon	The bond's fixed interest payment
Premium, discount bond	A bond priced above or below par
Callable, puttable bond	The issuer may repay early; the holder may demand early repayment
Convertible bond	A bond that can be exchanged for shares
Zero-coupon bond	Pays no coupons; bought at a discount and repaid at par
Covenant	A promise in a bond contract that protects lenders
Seniority	The order in which creditors are repaid in bankruptcy
Investment grade, high yield	Rated BBB-/Baa3 or better; rated below that ("junk")
Fallen angel	A bond downgraded from investment grade to high yield
Credit spread	Extra yield over a Treasury of the same maturity
Duration	A bond's sensitivity to interest rates, in years
Convexity	The curvature that makes price gains exceed losses for equal rate moves
Ladder, barbell, bullet	Maturities spread evenly; clustered at both ends; clustered around one date
Yield curve	Yields plotted against maturity
Inverted curve	Short-term yields above long-term yields; often a recession warning
Steepening, flattening	The gap between long and short yields widening or narrowing
Term premium	Extra yield demanded for lending longer
Real yield	Yield after inflation; TIPS quote it directly
Breakeven inflation	Nominal Treasury yield minus the TIPS yield of the same maturity
Flight to quality	Investors rushing into Treasuries and cash during turmoil
Reinvestment risk	The risk of reinvesting coupons or maturing bonds at lower rates

Economics

Term	Meaning
Recession	In NBER's words, a significant decline in economic activity spread across the economy and lasting more than a few months
Soft landing, hard landing	Inflation cooled without a recession; cooled by causing one
Stagflation	High inflation with weak growth and rising unemployment
Deflation, disinflation, reflation	Falling prices; slowing inflation; inflation picking back up
Hawkish, dovish	Leaning toward higher rates to fight inflation; toward lower rates to support growth
Tightening, easing	Raising rates or shrinking the balance sheet; cutting rates or expanding it
Neutral rate (r^*)	The real rate that neither speeds up nor slows down the economy
Output gap	Actual output minus potential output
Goldilocks	Growth neither too hot nor too cold, with tame inflation
Core inflation	Inflation excluding volatile food and energy
Sticky inflation	Prices, often services and wages, that adjust slowly
Base effects	Year-over-year rates distorted by unusual readings a year earlier
Wage-price spiral	Rising wages pushing up prices, which push up wages
Fiscal stimulus, austerity	Higher spending or tax cuts; spending cuts or tax rises
Debt ceiling	The legal cap on US federal borrowing
Trade deficit	Imports exceeding exports
Tariff	A tax on imports
Reserve currency	A currency held widely by central banks, chiefly the US dollar
Safe haven	An asset expected to hold value in a crisis, such as Treasuries or gold
Productivity	Output per hour worked; the engine of long-run growth

Portfolio management and risk

Term	Meaning
Asset allocation	How a portfolio is split among asset classes
Strategic vs tactical allocation	The long-term policy mix vs short-term tilts within ranges
Rebalancing, drift	Restoring target weights; weights wandering as prices move
Glide path	A planned shift toward safer assets as a goal approaches
Diversification	Spreading money across assets that do not move together
Correlation	How closely two assets move together, from -1 to $+1$
Efficient frontier	Portfolios with the highest expected return for each level of risk

Term	Meaning
Risk budget	The amount of risk allotted to each part of a portfolio
Risk parity	Weighting assets so each contributes equal risk
Alpha	Return beyond what the risk taken explains
Beta	Sensitivity to market moves
Tracking error	Volatility of returns relative to the benchmark
Active share	How much a portfolio's holdings differ from its benchmark
Home bias	Overweighting one's own country
Sequence - of - returns risk	Damage from poor returns early in a withdrawal period
Longevity risk	Outliving one's money
Shortfall risk	The chance of missing a goal
Tail risk, black swan, fat tails	Rare extreme losses; an unforeseeable shock; extremes more common than a bell curve predicts
Drawdown	A decline from a previous peak
Mean reversion	The tendency of values to drift back toward an average
Survivorship bias	Overstated results because failures dropped out of the data
Look - ahead bias	Using information that was not available at the time
Overfitting	A model that fits past noise rather than a real pattern
Stress test	Estimating losses in a defined severe scenario
Monte Carlo simulation	Thousands of random paths used to estimate the odds of outcomes
Hedge	A position that offsets the risk of another
Capital preservation	Protecting the original investment as the first priority
Total return approach	Spending from income and gains together, rather than income only
Bucket strategy	Dividing money by time horizon: near - term cash, medium - term bonds, long - term stocks

Companies, accounting and corporate finance

Term	Meaning
Top line, bottom line	Revenue; net income
Accrual accounting	Recording revenue and costs when earned or incurred, not when cash moves
Depreciation, amortization	Spreading the cost of physical and intangible assets over their lives
Goodwill	The premium paid in an acquisition over the fair value of net assets
Impairment	A write - down when an asset is worth less than its book value
Working capital	Current assets minus current liabilities
Operating leverage	High fixed costs that make profits swing more than sales
Guidance	Management's own forecast
Consensus estimate	The average analyst forecast
Beat, miss	Results above or below consensus
Non - GAAP (adjusted) earnings	Earnings that exclude items management considers unusual; read the reconciliation
Restatement	Correcting previously published financial statements; a red flag
Spin - off	Separating a division into an independent listed company
Moat	A durable competitive advantage
Synergy	Value from combining two companies beyond their separate values
Accretive, dilutive	A deal that raises or lowers earnings per share
Hurdle rate	The minimum return a project must earn
Economic profit	Profit above the cost of all capital: $(ROIC - WACC) \times \text{invested capital}$

Behavioral finance

Term	Meaning
Prospect theory	People judge outcomes as gains or losses from a reference point and weigh losses more
Loss aversion	Losses hurt roughly twice as much as equal gains feel good
Anchoring	Fixating on an initial number such as a purchase price
Confirmation bias	Seeking information that supports existing beliefs
Overconfidence	Overrating one's skill or knowledge
Recency bias	Overweighting recent events
Herding	Following the crowd
Disposition effect	Selling winners too soon and holding losers too long

Term	Meaning
Mental accounting	Treating money differently depending on its source or label
Framing	Decisions changing with how the same facts are presented
Status quo bias, regret aversion	Preferring inaction; avoiding decisions that might later be regretted
Hindsight bias	Believing past events were predictable
Availability bias	Judging likelihood by how easily examples come to mind

Wealth management, regulation and ethics

Term	Meaning
Fiduciary duty	The legal duty to act in the client's best interest, putting the client first
Suitability	Recommendations must fit the client's profile
Regulation Best Interest	SEC rule requiring broker - dealers to act in retail clients' best interest when recommending
Know your customer	Verifying identity and understanding the client before advising
Discretionary vs advisory mandate	The manager trades without prior approval; or recommends and the client decides
Custodian	The institution that holds client assets
Trust (revocable, irrevocable)	A legal arrangement holding assets for beneficiaries; changeable or fixed
Estate planning	Arranging the transfer of wealth: will, trusts, beneficiaries, powers of attorney
Step - up in basis	Heirs' cost basis resets to market value at death under current US law
Family office	A firm managing a wealthy family's investments and affairs
Accredited investor	Meets SEC wealth or income tests to buy private offerings
Fee - only vs commission - based adviser	Paid only by the client; paid by product sales, which can create conflicts
Wrap fee	One fee covering advice, trading and custody
Robo - adviser	Automated, algorithm - driven portfolio management
Prudent investor rule	Standard requiring trustees to invest with care, skill and diversification
CFA Institute Code and Standards	Ethics rules for investment professionals: integrity, objectivity, loyalty to clients
Asset Manager Code	CFA Institute's code for asset managers, which competition teams must follow
Insider trading	Trading on material non - public information; illegal
Front - running	Trading ahead of a client's order for personal gain; prohibited
Churning	Excessive trading to generate commissions; prohibited
Conflict of interest	Any situation where the adviser's interests could compete with the client's; must be disclosed and managed

Sources

- [InvestmentNews: IRS reveals new 401\(k\), IRA contribution limits for 2026](#): the 2026 limits in I17.
- [Damodaran Online: historical returns, 1928–2023](#): the US stock Sharpe ratio (D1) and losing-year count (K2), computed from its annual data.
- [Hartford Funds: S&P 500 bear markets](#): bear-market frequency (K2).

Formulas follow standard finance and economics texts and the CFA curriculum. Tax rules and thresholds change; confirm current figures with the IRS before relying on them.